

Uber Ride Bookings Analytics Dashboard — Project Breakdown & Interview Prep

Generated from a direct technical analysis of Uber.pbix (Power Query/M code, DAX measures, data model, and report layout extracted from the file itself).

1. One-line project summary (for resume / elevator pitch)

Built an end-to-end Power BI dashboard analyzing 150,000 ride-booking records for a Delhi-NCR ride-hailing service, covering booking volume, revenue, cancellations, and service ratings across 6 interactive report pages with custom navigation.

2. Project at a glance

Aspect	Detail
Data source	<code>ncr_ride_bookings.csv</code> (loaded via <code>File.Contents</code> , single CSV)
Rows / Columns	150,000 rows × 21 columns
Time period	Jan 1, 2024 - Dec 30, 2024
Data model	Single flat fact table (no star schema, no relationships)
Tables in model	<code>ncr_ride_bookings</code> + 2 Power BI auto-generated date tables
DAX measures	3 (<code>TotalBooking</code> , <code>CanceledBookings</code> , <code>CanceledPercentage</code>)
Report pages	6 (Home, Overall, Vehicle Type, Revenue, Cancellation, Ratings)
Navigation	Custom button-based navigation (PageNavigation action buttons), not default page tabs
Theme	Power BI built-in theme "CY24SU10" (default, not custom-built)
Row-level security	None implemented

Aspect	Detail
Likely source dataset	Matches Kaggle's "Uber Ride Analytics Dashboard" dataset structure (Booking ID format <code>CNR#####</code> , Customer ID format <code>CID#####</code>) — confirm against your own download history

3. The dataset, in detail

Columns in `ncr_ride_bookings`:

`(Date, Time, Booking ID, Booking Status, Customer ID, Vehicle Type, Pickup Location, Drop Location, Avg VTAT, Avg CTAT, Cancelled Rides by Customer, Reason for cancelling by Customer, Cancelled Rides by Driver, Driver Cancellation Reason, Incomplete Rides, Incomplete Rides Reason, Booking Value, Ride Distance, Driver Ratings, Customer Rating, Payment Method)`

Booking Status breakdown (150,000 total):

- Completed: 93,000 (62%)
- Cancelled by Driver: 27,000 (18%)
- Cancelled by Customer: 10,500 (7%)
- No Driver Found: 10,500 (7%)
- Incomplete: 9,000 (6%)

Vehicle Type breakdown: Auto (37,419) > Go Mini (29,806) > Go Sedan (27,141) > Bike (22,517) > Premier Sedan (18,111) > eBike (10,557) > Uber XL (4,449)

Payment Method: UPI is the leading method (45,909), followed by Cash (25,367), Uber Wallet (12,276), Credit Card (10,209), Debit Card (8,239). ~48,000 rows have no payment method (rides that never completed, so no payment was ever taken).

Cancellation reasons (customer side): Wrong Address, Change of plans, Driver is not moving towards pickup location, Driver asked to cancel, AC is not working. **Cancellation reasons (driver side):** Customer related issue, Customer was coughing/sick, Personal & car related issues, More than permitted people in vehicle. **Incomplete ride reasons:** Customer Demand, Vehicle Breakdown, Other Issue.

Key averages: Booking Value ≈ ₹508 avg, Driver Rating ≈ 4.23/5, Customer Rating ≈ 4.40/5, Ride Distance ≈ 24.6 km avg. ~177 distinct pickup locations, ~149,000 distinct customers (i.e., almost every booking is a different customer — low repeat-customer density in this dataset).

4. Data preparation (Power Query / M) — what you actually did

The full M code for the only query in the model:

```
m
let
    Source = Csv.Document(File.Contents("C:\Projects\Uber Booking\ncr_ride_bookings.csv",
        [Delimiter=",", Columns=21, Encoding=1252, QuoteStyle=QuoteStyle.None]),
        #"Promoted Headers" = Table.PromoteHeaders(Source, [PromoteAllScalars=true]),
        #"Changed Type" = Table.TransformColumnTypes(#"Promoted Headers",
            {{"Date", type date}, {"Time", type time}, {"Booking ID", type text},
            {"Booking Status", type text}, {"Customer ID", type text}, {"Vehicle Type", type text},
            {"Pickup Location", type text}, {"Drop Location", type text}, {"Avg VTAT", type number},
            {"Avg CTAT", type number}, {"Cancelled Rides by Customer", Int64.Type},
            {"Reason for cancelling by Customer", type text}, {"Cancelled Rides by Driver", Int64.Type},
            {"Driver Cancellation Reason", type text}, {"Incomplete Rides", Int64.Type},
            {"Incomplete Rides Reason", type text}, {"Booking Value", Int64.Type},
            {"Ride Distance", type number}, {"Driver Ratings", type number},
            {"Customer Rating", type number}, {"Payment Method", type text}})
in
    #"Changed Type"
```

What this tells an interviewer about your skill level honestly: the transformation logic is intentionally simple — header promotion and type casting only. There's no merging of tables, no unpivoting, no custom M functions, and no calculated columns at the Power Query stage. That's fine to say plainly: the source data was already fairly clean and tabular, so the main ETL work was correct type assignment (this matters more than it sounds — getting Date/Time/Int64/decimal types right up front is what makes DAX time intelligence and aggregations work correctly downstream).

5. Data modeling

- **Single-table model.** There is no separate dimension table for Vehicle Type, Payment Method, or Location — everything lives in one wide fact table. This means there are **no relationships** in the model.
- **Auto date tables.** Power BI's "Auto date/time" setting generated two hidden calendar tables (`DateTableTemplate_...` and `LocalDateTable_...`) — these were not built manually with `CALENDAR()` or `CALENDARAUTO()`, they're the built-in feature firing because a Date column exists.
- **No RLS, no model parameters, no calculation groups, no perspectives.**

This is worth being upfront about in an interview rather than overselling it — see Section 8 for how to frame this as a strength (you can speak to *why* a star schema matters even though this particular project didn't need one at this scale).

6. DAX measures — full code and what each one does

dax

```
TotalBooking = COUNTROWS(ncr_ride_bookings)
```

```
CanceledBookings =
```

```
CALCULATE(
```

```
    COUNTROWS(ncr_ride_bookings),
```

```
    ncr_ride_bookings[Booking Status] IN {"Cancelled by Driver", "Cancelled by Customer"
```

```
)
```

```
CanceledPercentage = DIVIDE([CanceledBookings], [TotalBooking], 0) * 100
```

- `TotalBooking`: a basic row-count measure used as the denominator for rate calculations and the headline KPI card.
- `CanceledBookings`: uses `CALCULATE` with an `IN` filter to count only rows where the booking was cancelled by either party (deliberately excludes "No Driver Found" and "Incomplete" — worth knowing why if asked: those are operationally different failure modes from an explicit cancellation).
- `CanceledPercentage`: wraps the two measures in `DIVIDE` with a 0 fallback (avoids divide-by-zero errors when a filter context returns no rows) and multiplies by 100 to express it as a percentage value rather than a decimal.

All the other "metrics" you see on the report (Sum of Booking Value, Sum of Ride Distance, Sum of Driver/Customer Ratings, counts via `CountNonNull`) are **implicit measures** — created automatically by dragging a numeric column into a visual's Values well and letting Power BI apply a default aggregation, rather than explicit DAX measures you wrote. That's a normal, common approach; just know the difference if asked "how many measures does your model have."

7. Report pages — what's on each one

Home Page — a landing page with action buttons (using Power BI's `PageNavigation` visual

link type) that route to each of the other 5 pages. No tab-based navigation; this was custom-built.

Overall — high-level KPIs: a line chart of booking volume over time (drillable through a Year → Quarter → Month → Day date hierarchy), a pie chart of Booking Status distribution, a date-range slicer, and a card visual.

Vehicle Type — a dense KPI-card grid: separate cards for Booking Value and Ride Distance broken out per vehicle type (Auto, Go Mini, Go Sedan, Premier Sedan, Uber XL, Bike/eBike), each driven by a visual-level filter on the `Vehicle Type` column rather than a single chart that handles all categories at once — this is a deliberate "card wall" design pattern.

Revenue — a column chart of Booking Value by Payment Method, a column chart of Ride Distance over the date hierarchy, and a table breaking down Booking Value by Customer ID.

Cancellation — two pie charts (Customer cancellation reasons, Driver cancellation reasons), KPI cards for total/completed/cancelled booking counts, and a card showing the `CanceledPercentage` measure.

Ratings — a card wall similar to the Vehicle Type page, showing average Customer Rating and Driver Ratings broken out per vehicle type.

Visual types used overall: line chart, pie chart, column chart, table (tableEx), card, slicer, action buttons, and one shape visual. No matrix, no scatter/map, no custom visuals from AppSource, no R/Python visuals.

8. Skills this project actually demonstrates (map these to your resume)

Skill	Evidence in the file
Power Query / M (ETL)	Wrote the M query for CSV ingestion, header promotion, and explicit type casting across 21 columns
DAX	Wrote 3 explicit measures using <code>CALCULATE</code> , <code>COUNTROWS</code> , <code>IN</code> , and <code>DIVIDE</code> with safe-division handling
Data modeling	Worked with Power BI's Tabular/VertiPaq model; understands implicit vs. explicit measures and auto-generated date tables
Data visualization	Selected appropriate chart types per question (trend → line, composition → pie, comparison → column, ranking → table) across 6 themed pages

Skill	Evidence in the file
Dashboard UX / navigation design	Built a custom button-driven navigation system instead of relying on default page tabs
Business/domain analysis	Translated raw booking-status and cancellation-reason data into operational KPIs (cancellation rate, revenue by payment method, performance by vehicle type)

Resume bullet suggestions (pick 2-3, tailor to role):

- "Built a 6-page Power BI dashboard analyzing 150K+ ride-booking records, using Power Query (M) for ETL and DAX (CALCULATE, DIVIDE) to surface cancellation rate, revenue, and rating KPIs."
- "Designed a custom button-based navigation system in Power BI, improving end-user experience over default page-tab navigation."
- "Analyzed ride-hailing operational data to identify that ~25% of bookings were lost to cancellations or driver no-shows, broken down by customer- vs. driver-initiated cancellation reasons."

9. Likely interview questions, with answers grounded in your actual file

"Walk me through your project."

Use the one-liner from Section 1, then go: data source → cleaning in Power Query → 3 DAX measures → 6-page report → one specific insight (e.g., "62% of bookings completed, but driver-initiated cancellations were almost 3x customer-initiated ones, which was the most interesting finding because it points to a supply-side problem, not a demand-side one").

"What transformations did you do in Power Query?"

Be honest and confident: header promotion and explicit type casting on all 21 columns (Date, Time, Int64, decimal, text). If they push for more — "did you do anything more complex, like merges or unpivots?" — say no, the source was already a clean flat file, and explain that getting types right (especially Date/Time split into two separate columns rather than one DateTime) was the main correctness issue to solve, since wrong types silently break time-intelligence functions and aggregations later.

"Walk me through one of your DAX measures."

Use `CanceledPercentage`. Explain: it depends on two other measures (`CanceledBookings` / `TotalBooking`), `CALCULATE` is used to apply a row filter inside `CanceledBookings` via an `IN`

condition, and `DIVIDE` is used instead of the `/` operator specifically to avoid a divide-by-zero error when a filter context (e.g., a slicer selection) returns zero total bookings — `DIVIDE`'s third argument supplies a safe fallback (0) instead of throwing an error.

"Why didn't you build a star schema / use separate dimension tables?"

Good honest answer: at this scale (one flat table, 150K rows, no need to reuse dimension values across multiple fact tables), a single-table model is a legitimate and common choice — star schemas earn their complexity when you have multiple fact tables that need to share dimensions, or when a dimension table needs its own attributes beyond what's in the fact table. If asked how you'd improve it: split `Vehicle Type`, `Payment Method`, and `Pickup/Drop Location` into separate dimension tables to reduce column cardinality in the fact table and make slicers/relationships explicit instead of relying on direct column filtering, which would also reduce the model's memory footprint at larger scale.

"What's the difference between an implicit and explicit measure, and which did you use?"

Explicit measures are DAX you write yourself and that live in the model independent of any one visual (your 3 measures). Implicit measures are created automatically when you drag a numeric column into a visual and Power BI applies a default aggregation (Sum, Average, etc.) — most of the "Sum of Booking Value" and "Sum of Ride Distance" cards in this report are implicit measures. Best practice in production models is to convert frequently-used implicit aggregations into explicit measures for consistency and reusability — a fair thing to mention as a "next step" if asked how you'd harden this for a real team.

"How would you make this dashboard production-ready / what would you improve?"

Strong, self-aware answers based on what's actually in the file:

- Replace the hardcoded local file path (`C:\Projects\Uber Booking\...`) with a parameter or a shared data source, so the report doesn't break when published or moved to another machine.
- Convert recurring implicit measures (Sum of Booking Value, Sum of Ride Distance, average ratings) into explicit DAX measures for governance and reuse.
- Consider splitting into a star schema if the model grows to include more fact tables.
- Turn off "Auto date/time" and build an explicit `CALENDAR()`-based date table if you need custom fiscal periods or want to reduce model size (auto date tables create one hidden table per date column, which adds up).
- Add a refresh schedule if this were connected to a live/growing data source rather than a static CSV snapshot.

"How did you decide which chart type to use for which page?"

Trend over time → line chart (Overall, Revenue pages); part-to-whole composition → pie chart (Booking Status, cancellation reasons); categorical comparison → column chart (Revenue by Payment Method); ranking/detail lookup → table (Customer ID by Booking Value); single headline numbers → cards (KPIs, broken out per vehicle type on dedicated pages).

"What was the hardest part of this project?"

Honest options grounded in the file: handling the high proportion of null values correctly (e.g., ~93% of Reason for cancelling by Customer is null because only cancelled bookings have a reason — meaning any chart needs to implicitly or explicitly filter to the relevant subset rather than showing nulls as a category), or designing a card-wall layout that breaks one KPI out across 7 vehicle types cleanly using visual-level filters instead of a single chart.

10. A note on honesty in interviews

Everything above is generated from your actual file — the M code, DAX, and report structure are exactly what's there, not invented. If you genuinely made the changes referenced (you said you "taken this project and made changes into it"), be ready to speak to *which specific changes were yours* versus what came with the original template, since an interviewer who knows Power BI may ask you to defend specific design choices line by line.